

**Question 1** (12 marks) **Marks**

- a) Solve  $\frac{3}{x-1} < 4$ . **3**
- b) Find  $\lim_{x \rightarrow 0} \frac{\sin x}{5x}$ . **1**
- c) Given  $\sin \theta + \cos \theta = A$  and  $\sin \theta \cos \theta = B$ . Express  $\sin^3 \theta + \cos^3 \theta$  in terms of  $A$  and  $B$ . **2**
- d) Find  $\int \frac{\cos \theta}{5 - \sin \theta} d\theta$  **2**
- e) The two parabolas  $y = x^2$  and  $y = (x-2)^2$  intersect at the point  $(1,1)$ . Find the angle between the tangents to the curves at  $x = 1$ . Give your answer to the nearest degree. **4**

**Question 2** (12 marks) Start this question in a new booklet.

- a) Use the substitution  $u = 9 - x^2$  to find  $\int_0^1 6x\sqrt{9-x^2} dx$ . **4**
- b) Given  $\frac{d}{dx}(x \sin x) = \sin x + x \cos x$ , find  $\int x \cos x dx$ . **2**
- c) (i) State the domain and range of the function  $y = 3 \sin^{-1}\left(\frac{x}{2}\right)$  **2**  
 (ii) Draw a sketch of the function, carefully labeling the extremities of both the range and domain. **1**
- d) The letters of the word *CALCULUS* are arranged in a row. **3**  
 (i) How many different arrangements are there?  
 (ii) If one of these arrangements is selected at random, find the probability that it begins with "U" and ends with a "U".

**Question 3** (12 marks) Start this question in a new booklet. **Marks**

- a) Consider the parabola  $x^2 = 4ay$  where  $a > 0$ , and suppose the tangents at  $P(2ap, ap^2)$  and  $Q(2aq, aq^2)$  intersect at the point  $T$ .
- (i) Given the equation of the tangent from  $P$  is  $y = px - ap^2$ , **2**  
show that the point  $T$  has coordinates,  $(a(p + q), apq)$
- (ii) Given  $S$  is the focus, show that  $SP = a(p^2 + 1)$ . **2**
- (iii) If  $P$  and  $Q$  move along the parabola such that  $SP + SQ = 4a$ , **2**  
find the equation of the locus of point  $T$ .
- b) The function  $f(x) = x - e^{-x}$  has one root between 0 and 1.
- (i) Show that the root lies between 0.4 and 0.6. **1**
- (ii) Use one application of Newton's Method with an **2**  
initial approximation of  $x = 0.5$  to find a second approximation.
- c) A polynomial  $f(x)$  leaves a remainder of  $-10$  when divided by **3**  
 $(x - 1)$  and  $20$  when divided by  $(x + 2)$ . If  $f(x) = x^n + ax - 6$ , show that  $x + 1$  is a factor.

**Question 4** (12 marks) Start this question in a new booklet.

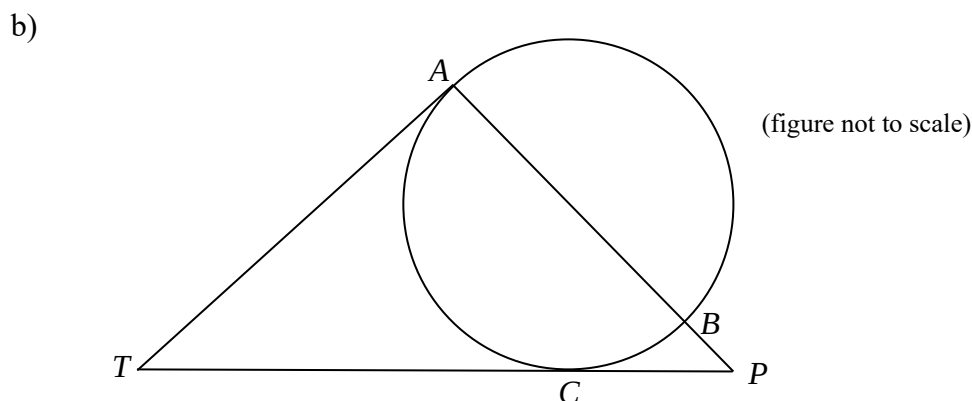
- a) The speed  $v$  cm/s of a particle with simple harmonic motion in a straight line is given by  $v^2 = 6 + 4x - 2x^2$ , where  $x$  cm is the magnitude of the displacement from a fixed point  $O$ .
- (i) Show that  $\ddot{x} = -2(x - 1)$ . **2**
- (ii) Find the centre of the motion. **1**
- (iii) Find the period of the motion. **1**
- (iv) Find the amplitude of the motion. **2**

**Question 4 continued****Marks**

- b) Find, as a rational number, the coefficient of  $x$  in the expansion of  $\left(x^2 + \frac{1}{2x}\right)^8$ . **3**
- c) Colour blindness affects 5% of all men. What is the probability that any random sample of 20 men should contain:
- (i) no colour blind men. **1**
- (ii) exactly 4 colour blind men. **2**

**Question 5** (12 marks) Start this question in a new booklet.

- a) (i) Show that  $\int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \cos^2 x \, dx = \frac{\pi}{8} - \frac{1}{4}$ . **2**
- (ii) Hence or otherwise find the volume generated when  $y = 1 - \cos x$  is rotated about the  $x$  axis between  $x = \frac{\pi}{4}$  and  $x = \frac{\pi}{2}$ . **2**



$AB$  is a diameter of a circle  $ABC$ . The tangents at  $A$  and  $C$  meet at  $T$ . The lines  $TC$  and  $AB$  are produced to meet at  $P$ .

Copy the diagram into your examination booklet. Join  $AC$  and  $CB$ .

- (i) Prove that  $\angle CAT = 90^\circ - \angle BCP$ . **2**
- (ii) Hence, or otherwise prove that  $\angle ATC = 2\angle BCP$ . **2**

**Question 5 continued****Marks**

- c) At time  $t$  the temperature  $T^0$  of a body in a room of constant temperature  $20^0$  is decreasing according to the equation  $\frac{dT}{dt} = -k(T - 20)$  for some constant  $k > 0$ .
- (i) Verify that  $T = 20 + Ae^{-kt}$ , where  $A$  is a constant, is a solution of the equation. **1**
- (ii) The initial temperature of the body is  $90^0$  and it falls to  $70^0$  after 10 minutes. Find the temperature of the body after a further 5 minutes. ( Answer to the nearest degree.) **3**

**Question 6** (12 marks) Start this question in a new booklet.

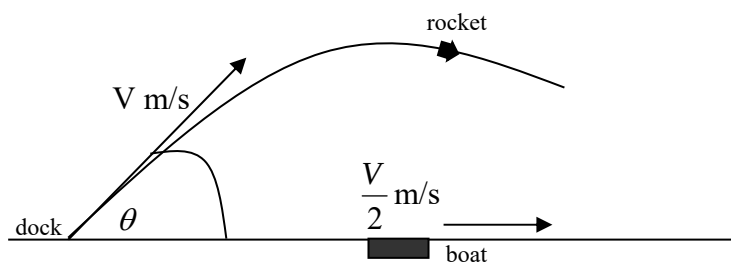
- a) Consider the function  $f(x) = \frac{e^x}{1 + e^x}$ .
- (i) Show that  $f(x)$  is increasing for all values of  $x$ . **2**
- (ii) Explain why  $f(x)$  has an inverse function. **1**
- (iii) Find the inverse function  $y = f^{-1}(x)$ . **2**
- b) A yacht is sailing due north off the west coast of a small island with an extinct volcanic peak of height 1200m.  
At 0900 hours the bearing of the peak is  $010^0$  and the angle of elevation  $5^0$ . At 1030 hours the bearing of the peak is  $120^0$  and the angle of elevation  $20^0$ .
- (i) Draw a diagram to illustrate the information above. **1**
- (ii) Calculate their speed in km/h. **3**
- c) (i) Express  $\sin x + \sqrt{3} \cos x$  in the form  $R \sin(x + \alpha)$ . **1**
- (ii) Hence solve  $\sin x + \sqrt{3} \cos x = 1$  for  $0 \leq x \leq 2\pi$  **2**

**Question 7** (12 marks) Start this question in a new booklet.**Marks**

- a) A boat leaves a dock at a speed of  $\frac{V}{2}$  m/s and travels in a straight line. A rocket is fired simultaneously from the same starting point and in the same direction, with initial speed  $V$  m/s and angle of elevation  $\theta$ , where  $0^\circ < \theta < 90^\circ$ . Also assume the height of the dock and the boat is zero metres.

Given the rocket's trajectory is defined by the equations:

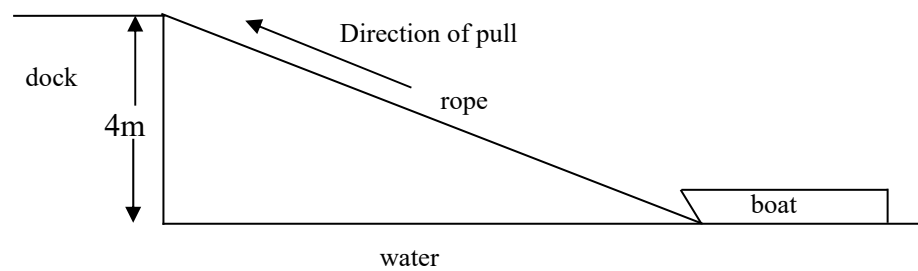
$$x = Vt \cos \theta \text{ and } y = -\frac{1}{2}gt^2 + Vt \sin \theta$$



- (i) show that the time of flight of the rocket is 2  
 $\frac{2V \sin \theta}{g}$  seconds.
- (ii) show that the range of the rocket is 1  
 $\frac{2V^2 \sin \theta \cos \theta}{g}$  metres.
- (iii) determine the angle of elevation of the rocket needed 2  
 for the rocket to hit the boat
- b) Use mathematical induction to prove that, for every positive 3  
 integer  $n$ ,  $9^n - 3$  is divisible by 6.

**Question 7 continued****Marks**

- c) A boat is pulled towards a dock by a rope through a ring on the dock which is 4 metres above the boat. If the rope is pulled in at 2 m/s, how fast is the boat travelling when the rope is 10 metres long? **4**

**End of paper.**

① (a)  $\frac{3}{x-1} \times (x-1)^2 < 4(x-1)^2$

$$3(x-1) < 4(x-1)^2$$

$$4(x-1)^2 - 3(x-1) > 0$$

$$(x-1)(4x-7) > 0$$

$$x < 1 \text{ or } x > \frac{7}{4}$$

(b)  $\frac{1}{5}$

(c)  $\sin^3 \theta + \cos^3 \theta$

$$= (\sin \theta + \cos \theta)(\sin^2 \theta - \sin \theta \cos \theta + \cos^2 \theta)$$

$$= A(1-B)$$

or

$$(\sin \theta + \cos \theta)^3 = \sin^3 \theta + 3\sin^2 \theta \cos \theta + 3\sin \theta \cos^2 \theta + \cos^3 \theta$$

$$\therefore \sin^3 \theta + \cos^3 \theta = (\sin \theta + \cos \theta)^3 + 3\sin \theta \cos \theta (\sin \theta + \cos \theta)$$
$$= A^3 - 3BA$$

(d)  $-\ln(5 - \sin \theta) + C$

(e) Let  $\theta$  = acute angle between lines

$$\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$$

$$m_1 = 2x, \text{ at } x=1$$
$$= 2$$

$$m_2 = 2(x-2) \text{ at } x=1$$
$$= -2$$

$$\tan \theta = \left| \frac{2 - (-2)}{1 + 2 \cdot (-2)} \right|$$

$$= \left| -\frac{4}{3} \right|$$

$$\theta = 53^\circ \text{ (nearest degree)}$$

$$2(a) \int_0^1 6x\sqrt{9-x^2} dx$$

$$= -3 \int_0^1 \sqrt{9-x^2} (-2x dx)$$

$$= -3 \int_9^8 u^{1/2} du$$

$$= -3 \left[ \frac{2}{3} u^{3/2} \right]_9^8$$

$$= -3 \left( \frac{2}{3} \sqrt{512} - 18 \right)$$

$$= -2\sqrt{512} + 54$$

$$= -32\sqrt{2} + 54$$

$$u = 9 - x^2$$

$$\therefore du = -2x dx$$

$$\text{if } x=1, u=8$$

$$x=0, u=9$$

$$8.745166004$$

$$(b) \frac{d}{dx} (x \sin x) = \sin x + x \cos x$$

$$\therefore x \cos x = \frac{d}{dx} (x \sin x) - \sin x$$

$$\therefore \int x \cos x dx = \int \left[ \frac{d}{dx} (x \sin x) - \sin x \right] dx$$

$$= x \sin x - \int \sin x dx$$

$$= x \sin x + \cos x + C$$

$$(c) (i) y = 3 \sin^{-1} \left( \frac{x}{2} \right)$$

$$\therefore \frac{1}{3} y = \sin^{-1} \left( \frac{x}{2} \right)$$

$$\therefore \sin \frac{1}{3} y = \frac{x}{2}$$

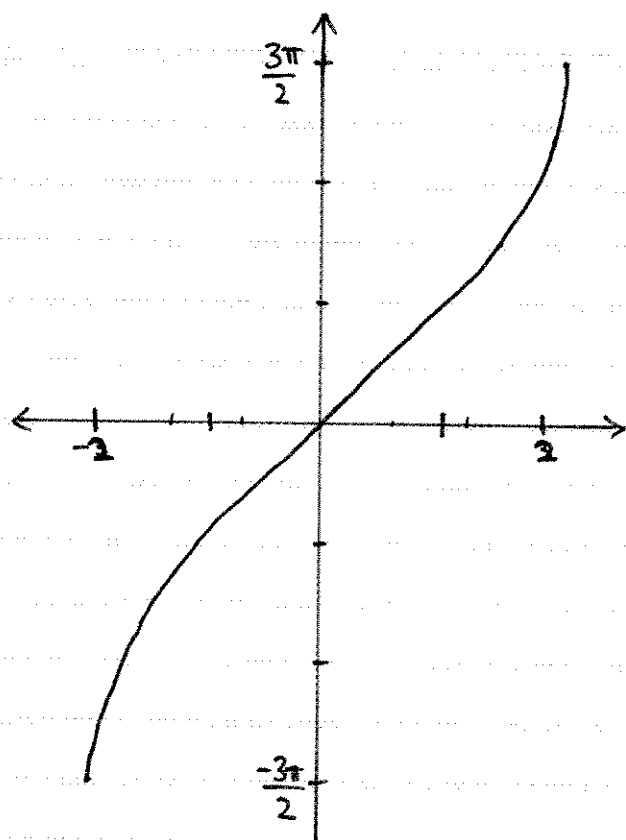
$$\therefore 2 \sin \frac{1}{3} y = x$$

$$\therefore -2 \leq x \leq 2$$

$$-\frac{3\pi}{2} \leq y \leq \frac{3\pi}{2}$$



(ii)



$$(d)(i) \text{ Arrangements} = \frac{8!}{2!2!2!} = 5040$$

$$(ii) \text{ Probability} = \frac{6!}{2!2!} = \frac{180}{5040}$$

$$= \frac{1}{28}$$

$$= \frac{1}{28}$$

for simpl. from factorial form

$$\text{OR Prob} = (\text{Prob } 1^{\text{st}} \text{ letter is a U}) \times (\text{Prob. last letter is a U})$$

$$= \frac{2}{8} \times \frac{1}{7}$$

$$= \frac{1}{28}$$

$$(3) (a) (i) y = px - ap^2 \text{ ——— } (1)$$

$$y = qx - aq^2 \text{ ——— } (2)$$

Solve (1), (2) simultaneously

$$x = a(p+q), y = apq$$

$$(ii) S(0, a), P(2ap, ap^2)$$

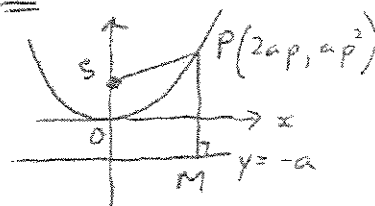
$$SP = \sqrt{(2ap-0)^2 + (ap^2-a)^2}$$

$$= \sqrt{4a^2p^2 + a^2p^4 - 2a^2p^2 + a^2}$$

$$= \sqrt{a^2(p^2+1)^2}$$

$$= a(p^2+1)$$

or



$$SP = PM \text{ (by definition)}$$

$$= ap^2 - (-a)$$

$$= ap^2 + a = a(p^2+1)$$

$$(iii) SP = a(p^2+1), SQ = a(q^2+1)$$

$$a(p^2+1) + a(q^2+1) = 4a$$

$$p^2 + q^2 = 2 \text{ ——— } (3)$$

$$x = a(p+q), y = apq$$

$$x^2 = a^2(p^2 + q^2 + 2pq)$$

$$= a^2\left(2 + \frac{2apq}{a}\right) \text{ from (3)}$$

$$= a^2\left(2 + \frac{2y}{a}\right)$$

$$= 2a(a + y)$$

$$(4) (i) f(0.4) < 0, f(0.6) > 0$$

or

$$f(0.4) \cdot f(0.6) < 0$$

$\therefore$  root exists

between 0.4 & 0.6

$$\begin{aligned}
 \text{b (ii)} \quad x_2 &= x_1 - \frac{f(x_1)}{f'(x_1)} \\
 &= x_1 - \frac{x_1 - e^{-x_1}}{1 + e^{-x_1}}
 \end{aligned}$$

$$x_1 = 0.5$$

$$\therefore x_2 \approx 0.57$$

$$(c) \quad f(1) = -10, \quad f(2) = 20$$

$$\therefore 1^n + a - 6 = -10$$

$$a = -5$$

$$(-2)^n + 10 - 6 = 20$$

$$(-2)^n = 16$$

$$n = 4$$

$$\therefore f(x) = x^4 - 5x - 6$$

$$f(-1) = 1 + 5 - 6$$

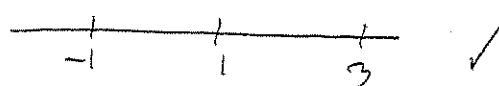
$$= 0$$

$\therefore x+1$  is a factor.

### Question 4

$$\begin{aligned}
 \text{a) i) } \ddot{x} &= \frac{d}{dt} \left( \frac{1}{2} v^2 \right) = \frac{d}{dt} \left( \frac{1}{2} (6 + 4x - 2x^2) \right) \checkmark \\
 &= \frac{d}{dt} (3 + 2x - x^2) \checkmark \\
 &= 2 - 2x \\
 &= -2(x-1)
 \end{aligned}$$

ii) clearly centre is  $x=1$ .  
 $V=0$  at end points  
 $\therefore 0 = 6 + 4x - 2x^2$   
 $0 = x^2 - 2x - 3$   
 $0 = (x-3)(x+1)$   
 $x = -1$  or  $3$



$\therefore$  centre of motion is  $x=1$

iii)  $n^2 = 2$   
 $n = \sqrt{2}$   
 Period  $= \frac{2\pi}{n}$   
 $= \frac{2\pi}{\sqrt{2}}$   
 $= \sqrt{2} \pi \checkmark$

iv) From part ii  
 amplitude  $= 2 \checkmark$

### Question 4

b)  $(x^2 + \frac{1}{2x})^8 = \sum_{r=0}^8 (x^2)^r \left(\frac{1}{2x}\right)^{8-r} {}^8C_r \checkmark$   
 $(x^2)^r \left(\frac{1}{2x}\right)^{8-r} = -x^1$   
 $2r + -1(8-r) = 1$   
 $2r - 8 + r = 1$   
 $3r = 9$   
 $r = 3 \checkmark$

Coefficient  $= {}^8C_3 \left(\frac{1}{2}\right)^{8-3}$   
 $= {}^8C_3 \left(\frac{1}{2}\right)^5$   
 $= 1.25 \checkmark$

c) i)  $(0.95)^{20} \doteq 0.36 \quad \cdot 35^{-8} \checkmark$

ii)  ${}^{20}C_4 (0.05)^4 (0.95)^{16} \doteq 0.013 \checkmark$

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Q5(a)(i)  $\int_{\pi/4}^{\pi/2} \cos^2 x \, dx = \frac{1}{2} \int_{\pi/4}^{\pi/2} (\cos 2x + 1) \, dx$  ✓

$= \frac{1}{2} \left[ \frac{1}{2} \sin 2x + x \right]_{\pi/4}^{\pi/2}$

$= \frac{1}{2} \left[ \left( \frac{1}{2} \sin \pi + \frac{\pi}{2} \right) - \left( \frac{1}{2} \sin \frac{\pi}{2} + \frac{\pi}{4} \right) \right]$  ✓

$= \frac{1}{2} \left( \frac{\pi}{2} - \frac{1}{2} - \frac{\pi}{4} \right)$

$= \frac{\pi}{8} - \frac{1}{4}$

(2) generally well done.

(ii) Vol =  $\pi \int_{\pi/4}^{\pi/2} (1 - \cos x)^2 \, dx$

$= \pi \int_{\pi/4}^{\pi/2} 1 - 2\cos x + \cos^2 x \, dx$  ✓

$= \pi \left[ x - 2\sin x + \frac{x}{2} - \frac{1}{4} \right]_{\pi/4}^{\pi/2}$

$= \pi \left( \left( \frac{\pi}{2} - 2 + \frac{\pi}{4} - \frac{1}{4} \right) - \left( \frac{\pi}{4} - \sqrt{2} + \frac{\pi}{8} - \frac{1}{4} \right) \right)$

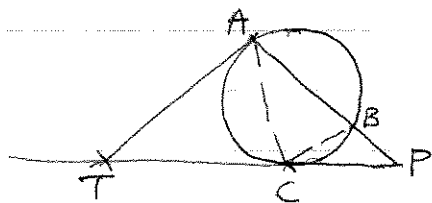
$= \pi \left( \frac{3\pi}{8} - 2\frac{1}{4} + \sqrt{2} \right)$  ✓

some did not

use part (i).

(2) Many messed up substitution.

Q5(b)



(i)  $\hat{BAC} = \hat{BCP}$  (alt segment theorem) ✓

$\hat{BAT} = 90^\circ$  (angle between tangent and diameter) ✓

$\therefore \hat{CAT} = \hat{BAT} - \hat{BAC}$

$= 90^\circ - \hat{BCP}$

many answers

lacked sequence

or had poor

(2) explanations.

(ii) Since  $AT = CT$  (tangents from external point to a circle are equal length) ✓

$\therefore \triangle ATC$  is isosceles.

So  $\hat{ACT} = \hat{CAT} = 90^\circ - \hat{BCP}$  (base  $\angle$  of  $\triangle$ ) ✓

$\therefore \hat{ATC} = 180^\circ - 2(90^\circ - \hat{BCP})$  ( $\angle$  sum  $\triangle$ ) ✓

$= 2\hat{BCP}$  //

(2)

5(c)(i)  $\frac{dT}{dt} = -kAe^{-kt}$

$= -k(T-20)$  (since  $Ae^{-kt} = T-20$ ) ✓ (1)

(ii)  $t=0$ ,  $T=90$  then  $A=70$  ✓

$T = 20 + 70e^{-kt}$

When  $T=70$ ,  $t=10$

$70 = 20 + 70e^{-10k} \Rightarrow k = -\frac{1}{10} \ln \frac{5}{7}$  ✓

When  $T=65$ ,  $T = 20 + 70e^{\frac{1}{10} \ln \frac{5}{7} \times 15}$

$= 62^\circ$  (nearest degree) ✓

(3)

well done

## Question 6

$$\begin{aligned} \text{a) i) } f'(x) &= \frac{e^x(1+e^x) - e^x \cdot e^x}{(1+e^x)^2} \\ &= \frac{e^x + e^{2x} - e^{2x}}{(1+e^x)^2} \\ &= \frac{e^x}{(1+e^x)^2} \end{aligned}$$

$> 0$  for all values of  $x$

$\therefore f(x)$  always has a positive gradient

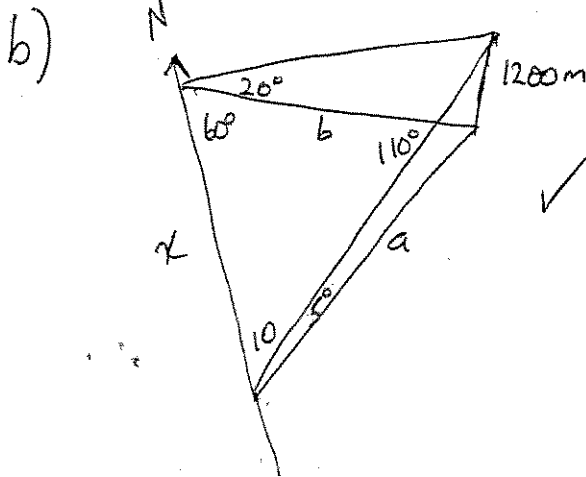
$\therefore f(x)$  is increasing for all values of  $x$

$$\begin{aligned} \text{iii) } x &= \frac{e^y}{1+e^y} \\ (1+e^y)x &= e^y \\ x + xe^y &= e^y \\ xe^y - e^y &= -x \end{aligned}$$

ii) Since  $f(x)$  is always increasing for each  $y$  value there can only be one  $x$  value.

$$\begin{aligned} e^y(x-1) &= -x \\ e^y &= \frac{-x}{x-1} \end{aligned}$$

$$y = \ln\left(\frac{-x}{x-1}\right)$$



$$\begin{aligned} \tan 5^\circ &= \frac{1200}{a} & \tan 20^\circ &= \frac{1200}{b} \\ a &= \frac{1200}{\tan 5^\circ} & b &= \frac{1200}{\tan 20^\circ} \end{aligned}$$

$$x^2 = \left(\frac{1200}{\tan 5^\circ}\right)^2 + \left(\frac{1200}{\tan 20^\circ}\right)^2 - 2\left(\frac{1200}{\tan 5^\circ}\right)\left(\frac{1200}{\tan 20^\circ}\right)\cos 110^\circ$$

$$x = 15163.56578 \text{ m}$$

$$\text{Speed} = \frac{15.163}{1\frac{1}{2}} \div 10.1 \text{ km/h}$$

c)  $\sin x + \sqrt{3} \cos x$

$$2\left(\frac{1}{2} \sin x + \frac{\sqrt{3}}{2} \cos x\right)$$

$$2(\cos 60^\circ \sin x + \sin 60^\circ \cos x)$$

$$2\left(\sin\left(x + \frac{\pi}{3}\right)\right)$$

$$\sin\left(x + \frac{\pi}{3}\right) \leq \frac{1}{2}$$

$$x + \frac{\pi}{3} \leq \frac{\pi}{6} \text{ or } \frac{5\pi}{6}$$

$$x = \frac{\pi}{6} \text{ or } \frac{\pi}{2}$$



7.(a)(i)  $y=0$  when rocket hits ground

$$\therefore -\frac{1}{2}gt^2 + Vt \sin \theta = 0$$

$$\therefore t(-\frac{1}{2}gt + V \sin \theta) = 0$$

$$\therefore \text{either } t=0 \text{ or } \frac{1}{2}gt = V \sin \theta$$
$$t = \frac{2V \sin \theta}{g}$$

(ii) when  $t = \frac{2V \sin \theta}{g}$  :

$$x = \frac{2V^2 \sin \theta \cos \theta}{g}$$

(1) ✓

(iii) for the boat:  $x = \frac{Vt}{2}$

$$\therefore x = \frac{2V^2 \sin \theta}{2g}$$

(2) ✓

Equating (1) and (2) :

$$\frac{2V^2 \sin \theta \cos \theta}{g} = \frac{2V^2 \sin \theta}{2g}$$

$$\therefore 2 \cos \theta = 1$$

$$\therefore \cos \theta = \frac{1}{2}$$

$$\therefore \theta = \frac{\pi}{3} \quad (\text{must be in first quadrant})$$

(b) To prove:  $9^n - 3$  is divisible by 6 for all positive integers  $n$

$$\text{For } n=1 : 9^1 - 3 = 6$$

$\therefore$  true for  $n=1$

Assume true for  $n=k$

$$\therefore 9^k - 3 = 6M \text{ where } M \text{ is an integer} \quad \checkmark$$

Prove true for  $n=k+1$

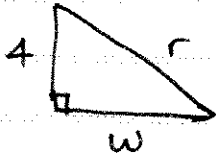
$$\begin{aligned} 9^{k+1} - 3 &= 9 \cdot 9^k - 3 \\ &= 9(6M + 3) - 3 \\ &= 54M + 27 - 3 \\ &= 54M + 24 \\ &= 6(9M + 4) \end{aligned} \quad \checkmark$$

$\therefore$  true for  $n=k+1$

Since it is true for  $n=1$ , and when we assume that it is true for  $n=k$  then it follows that it is also true for  $n=k+1$ , then it must be true for all integers greater than or equal to 1.



(c)



$$r^2 = 16 + w^2$$

$$r = \sqrt{16 + w^2}$$

✓

$$\therefore \frac{dr}{dw} = \frac{1}{2} \times 2w \times (16 + w^2)^{-1/2}$$

$$= \frac{w}{\sqrt{16 + w^2}}$$

$$\frac{dr}{dt} = 2$$

$$\begin{aligned} \therefore \frac{dw}{dt} &= \frac{dw}{dr} \times \frac{dr}{dt} \\ &= \frac{\sqrt{16 + w^2}}{w} \times 2 \end{aligned}$$

✓

when  $r = 10$ , ~~100~~  $100 = 16 + w^2$

$$\therefore w = \sqrt{84}$$

✓

$$\therefore \frac{dw}{dt} = \frac{\sqrt{16 + 84}}{\sqrt{84}} \times 2$$

$$= \frac{20}{\sqrt{84}}$$

✓

2.18